

Day 2: Network Topologies

Practical Answers

Q1. Bus Topology Diagram

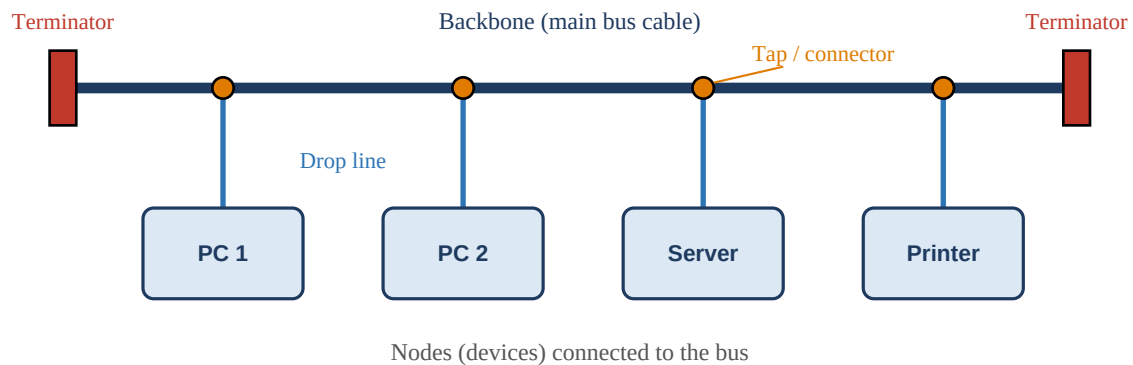


Figure 1: Bus topology with labelled components

Key components: the **backbone** is the single shared cable; **terminators** at both ends absorb the signal so it does not bounce back; **drop lines** and **taps/connectors** link each node to the backbone; **nodes** are the computers, server and printer.

Real-world example: Early Ethernet (10BASE2 / 10BASE5 coaxial cable) in small office or school computer labs used a bus layout, where all PCs shared one coaxial cable. Today it is mostly seen in small temporary networks and in industrial buses such as CAN bus in vehicles.

Q2. Star Topology in Cisco Packet Tracer

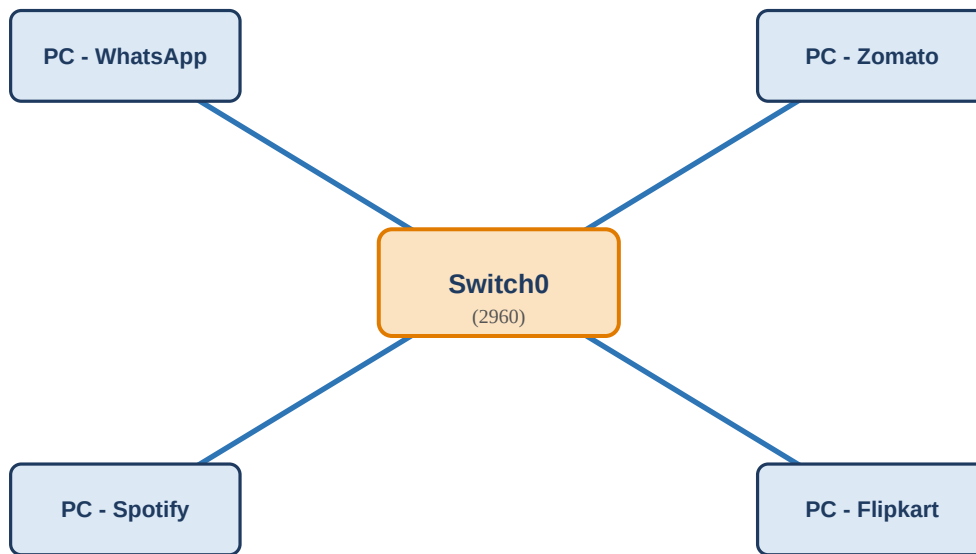


Figure 2: Star topology - 1 switch + 4 PCs named after popular apps

Steps to build it in Packet Tracer:

1. Open Packet Tracer, go to *End Devices* and drag 4 PCs onto the workspace.
2. Go to *Network Devices* > *Switches* and place one Switch (2960) in the centre.
3. Choose *Connections* > *Copper Straight-Through*. Click a PC (FastEthernet0), then the switch (FastEthernet0/1). Repeat for all 4 PCs on ports 0/2, 0/3, 0/4.
4. Click each PC label and rename it: WhatsApp, Zomato, Spotify, Flipkart.
5. (Optional) Assign IPs under *Desktop* > *IP Configuration*: 192.168.1.1 to .4 with mask 255.255.255.0, and test with ping.
6. Wait for all link lights to turn green, then take the screenshot (Windows: Win + Shift + S) and paste it here.

Note: the figure above is an illustration of the expected layout. Your own Packet Tracer screenshot should be inserted below.

[Paste your Packet Tracer screenshot here]

Q3. Ring Topology Diagram

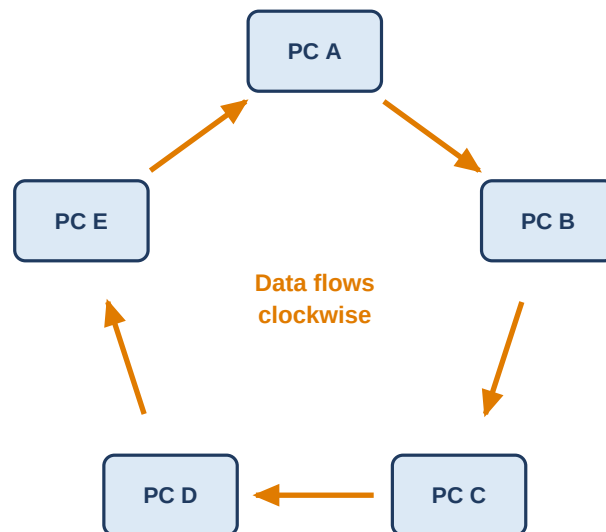


Figure 3: Ring topology - arrows show direction of data flow

How data travels: Each device is connected to exactly two neighbours, forming a closed loop. Data is sent as a packet (often with a token) in one direction only, e.g. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A$. Every device receives the packet and checks the destination address; if it is not for that device, it regenerates and passes it to the next device until it reaches the intended receiver.

Q4. Mesh vs Star Topology

Topology	Advantages	Disadvantages
Mesh	1. Very reliable: multiple paths exist, so if one link or device fails, data takes another route. 2. No traffic bottleneck: dedicated links carry data directly between devices, giving high speed.	1. Very expensive: needs many cables and ports ($n(n-1)/2$ links for n devices). 2. Complex to install and manage; adding a new device means connecting it to every other.
Star	1. Easy to install and expand: adding a device just needs one cable to the central switch. 2. Easy fault finding: a failed cable affects only that one device, not the whole network.	1. Single point of failure: if the central switch/hub fails, the entire network goes down. 2. Needs more cable than bus, and performance depends on the capacity of the central device.